MASTER EVIDENCE REGISTER

> **What this file is.** The central register: one ID per substantive claim, with the exact repository location that supports it, the verification status, and the limitation that keeps the claim honest. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Status vocabulary** (the six categories the brief requires): **IMPL** = VERIFIED AND IMPLEMENTED; **NOT IMPL** = VERIFIED BUT NOT IMPLEMENTED; **PASS** = TESTED AND PASSED; **FAIL** = TESTED AND FAILED; **NOT TESTED**; **UNKNOWN** = UNKNOWN / REQUIRES CONFIRMATION.
>
> **Evidence hierarchy applied:** code and schema first, then executed tests, then reproducible runs, then deployment logs, then version-controlled documentation, then drafts, and never a draft as proof of implementation.

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1. System structure and data

ID	Claim	Evidence	Status	Limitation
EV-01	The API is an Express application with 14 controllers and 12 route modules	`backend/src/index.js`; `backend/src/routes/*.js`; `backend/src/controllers/*.js`	IMPL	—
EV-02	The frontend is a React SPA with 14 page components, 3 shared components, an auth context and a hash-token hook	`frontend/src/pages/`, `frontend/src/components/`, `frontend/src/context/AuthContext.jsx`, `frontend/src/hooks/useHashToken.js`	IMPL	—
EV-03	The ML service is a separate Python package invoked as a subprocess from Node	`ml-service/predict.py --server`; `backend/src/services/duplicateCheck.js:262`	IMPL	Spawned per process; not containerised in the local path
EV-04	PostgreSQL holds 13 application tables	`backend/src/db/schema.sql`; `backend/src/db/migrations/`; schema doc §2	IMPL	Table count excludes the migration ledger
EV-05	Schema changes are forward-only, ledger-tracked and checksum-verified	14 forward migrations plus one `.down.sql` in `backend/src/db/migrations/`; `backend/src/services/databaseStartup.js`	IMPL	Not replayed against a fresh cluster in this pass
EV-06	Database readiness is checked before traffic is accepted	`backend/src/services/databaseReadiness.js`	IMPL	Readiness false-pass was a defect, now fixed (see §3)
EV-07	Payment state is gated by database constraints as well as API checks	schema doc §9; `disbursements` constraints	IMPL	Triggers inspected, not fuzzed

2. Behaviour

ID	Claim	Evidence	Status	Limitation
EV-08	Two sign-in endpoints issue a JWT; bcrypt hashes passwords	`backend/src/controllers/auth.controller.js`; `backend/src/config/jwtSecret.js`	IMPL	bcrypt cost not benchmarked on the host (NFR-02)
EV-09	Role is re-read from the database on each request, not trusted from the token	`backend/src/middleware/auth.js`; `requireRole.js`	IMPL	—
EV-10	Login and verification endpoints are rate limited	`loginRateLimit.js`, `verificationRateLimit.js`, `certificateRateLimit.js`	IMPL	Limits are per-process, so horizontal scale weakens them
EV-11	Registration validates fields with Zod inside the controller	`backend/src/controllers/students.controller.js`	IMPL	—
EV-12	Only five configured schools are accepted	`backend/src/config/schools.js`	IMPL	The five are a project assumption, not a ministry list

ID	Claim	Evidence	Status	Limitation
EV-13	Screening compares the submitted record with every registry record	`students.controller.js:75`; `mlDuplicateReview.js:84` (`scorePairs`)	IMPL	Quadratic; 22.67 ms measured at 50 pairs only
EV-14	Pair scores are aggregated to one candidate per record (`top-1`)	`backend/src/services/duplicateAggregation.js`	IMPL	Aggregation choice is a design decision, not validated against alternatives
EV-15	Model version is pinned for the duration of a scan	`duplicateFlags.controller.js:188–196`	IMPL	Guard raises on change; no retry logic
EV-16	ML failure degrades to manual review rather than a silent pass	`backend/src/services/duplicateCheck.js`	IMPL	Degradation path tested at route level only
EV-17	The duplicate queue masks identifiers; reveal is explicit and audited	`duplicateFlags.controller.js`; `duplicate_review_events`	IMPL	—
EV-18	Decisions require a reason and move the case to a terminal state	`duplicateFlags.controller.js`; state machine in schema doc §9	IMPL	—
EV-19	Eligibility is computed from five weighted factors into four categories	`backend/src/services/eligibility.js`, `bursaryRules.js`	IMPL	Arithmetic partially tested (FR-12 partial)
EV-20	Awards require verification evidence before approval	`bursaryAwards.controller.js`; `bursary_award_verification_events`	IMPL	—
EV-21	Disbursements support issue, clearance, officer-assisted acknowledgement and discrepancy detection	`disbursements.controller.js`; `disbursement_reconciliation_events`	IMPL	—
EV-22	QR tokens are 32 random bytes carried in the URL fragment	`verification.controller.js`; `frontend/src/hooks/useHashToken.js`	IMPL	Fragment handling stops token leakage into logs; it does not make the token secret to the holder
EV-23	Public verification asserts existence only	`verification.controller.js`; `verificationRoute.test.js`	PASS	Confirms existence, never identity, eligibility or payment
EV-24	Certificates store a snapshot, carry a reference and a token, and produce an A4 landscape PDF	`certificates.controller.js`; `certificateDocument.js`	IMPL	PDF content asserted at route level, not visually verified (FR-26 partial)
EV-25	Certificate state `not_current` is honoured when an award changes	`certificates.controller.js`; `bursary_certificates`	IMPL	—
EV-26	Audit rows are written for administrative and reveal actions	`backend/src/services/auditLog.js`	IMPL	Append-only by convention, not by constraint (NFR-21 partial)

3. Tests, defects and deployment

ID	Claim	Evidence	Status	Limitation
EV-27	The Jest suite passes 24 suites / 151 tests	`npx jest --runInBand tests`, 2026-11-01	PASS	One earlier sweep returned 149 or 150 of 151 under memory pressure (G-02)
EV-28	The native suite passes 43/43 in-process and 3/3 out-of-process	`node --test`, 2026-10-31	PASS	Separate layer; not additive with EV-27
EV-29	33 ML-service tests and 7 research-helper tests pass	`unittest discover`, 2026-10-31	PASS	Unit-level only
EV-30	Browser checks pass 23/23, 9/9 and 11/11	`scripts/browser-smoke.mjs`, `qr-smoke.mjs`, `registration-fields-smoke.mjs`, 2026-10-30 and 2026-11-01	PASS	A fourth harness, `admin-reveal-smoke.mjs`, was not re-run (admin password drifted) and is recorded as not run
EV-31	The frontend builds for production	`npm --prefix frontend run build`, exit 0, 2026-11-01	PASS	Build only; no bundle-size budget asserted

ID	Claim	Evidence	Status	Limitation
EV-32	A backup can be restored into an independent database with matching digests	`pg_dump` / `pg_restore`, 2026-10-31	PASS	One rehearsal on synthetic data (NFR-16 partial)
EV-33	Six defects were found in testing and fixed, each with a validating test	`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md` §5.4.7	PASS	Table is scoped to six; the wider history is in the handover and Azure logs
EV-34	A deployed instance answers health and registration-field checks	`docs/azure-deployment-log-2026-10-31.md`; `registration-fields-smoke.mjs`	PASS	Demonstration instance, not production
EV-35	The ML operating point is precision 0.9204, recall 0.8927, F1 0.9063 at a 0.50 cut-off, with a 7.72% hard-negative false-positive rate	`ml-service/evaluate_operating_point.py` output, transcribed in `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §7	PASS	**Synthetic data only**; pair-level, not beneficiary-level
EV-36	The independent per-pair baseline is 0.35% on 2 000 synthetic unrelated pairs	`docs/ml-evaluation-baseline-2026-10.md`	PASS	Answers a different question from EV-35; both must be quoted together
EV-37	The production model is XGBoost over six features, not the proposal's Random Forest over four	`ml-service/model/`; `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §5	PASS	Deviation must be stated in Chapters 4, 5 and 6 (G-21)
EV-38	A ten-feature Random Forest exists only as a research candidate	`ml-service/model/research/rf_attachment_style_candidate.joblib`	IMPL	Must never be presented as the production model
EV-39	Screening latency is 22.67 ms at 50 candidate pairs	`performance.test.js`	PASS	Nothing measured above 50 pairs; no load test exists
EV-40	The Docker image is built by CI, not locally	`.github/workflows/`; `Dockerfile`	IMPL	Engine unavailable on the development host, so no local build was observed

4. Claims with no supporting evidence, recorded as such

ID	Claim	Why it has no evidence	Status
EV-41	Real-world duplicate-detection accuracy	No adjudicated labels exist	NOT TESTED
EV-42	User acceptance and usability	No study, no participants, no consent record	NOT TESTED
EV-43	Behaviour under load or at scale	No load test was run	NOT TESTED
EV-44	A phone camera scanning a printed QR code	No device capture exists	NOT TESTED
EV-45	Penetration-test resistance	No assessment was commissioned	NOT TESTED
EV-46	Production uptime, monitoring or backup schedule	Demonstration instance only	NOT TESTED
EV-47	The exact Chapter 1 objective wording	Chapters 1–3 are not in this repository	UNKNOWN
EV-48	The wording of supervisor findings SBMS-01…SBMS-18	Unrecoverable; `SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md` §4 records the loss	UNKNOWN
EV-49	Whether the deployment's ML path is containerised or subprocess-spawned at runtime	Azure configuration shows the app service; the local path spawns Python	UNKNOWN (G-22)
EV-50	Whether `docs/verification.md` and `docs/uat-test-plan.md` remain part of the evidence base	Both conflict with the implementation; the decision is the author's	UNKNOWN (G-07, G-08)
EV-51	Whether the hardware specification can be reconstructed	Never recorded	UNKNOWN (G-01)

5. How to cite an ID

In a chapter: "(EV-27, `MASTER_EVIDENCE_REGISTER.md`)" or, where a section is more useful, the section reference the row already carries. Two rules:

1. **An ID never travels alone.** Any claim quoted from this register must keep the limitation column with it; the limitation is the part that makes the claim defensible.
2. **No new ID without a location.** If a sentence in a draft cannot be tied to a row here, either the sentence is wrong or a row is missing; it must not be written and left unsupported.

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6. Related files

`EVIDENCE_AND_VERIFICATION_REPORT.md` §2–§6 (claim-level index in prose), `SBMS_TEST_EVIDENCE_CATALOGUE.md` §2 (every test artefact with date and command), `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §4 (the forbidden claims), `ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md` §3 (the same gaps in academic-coverage form).